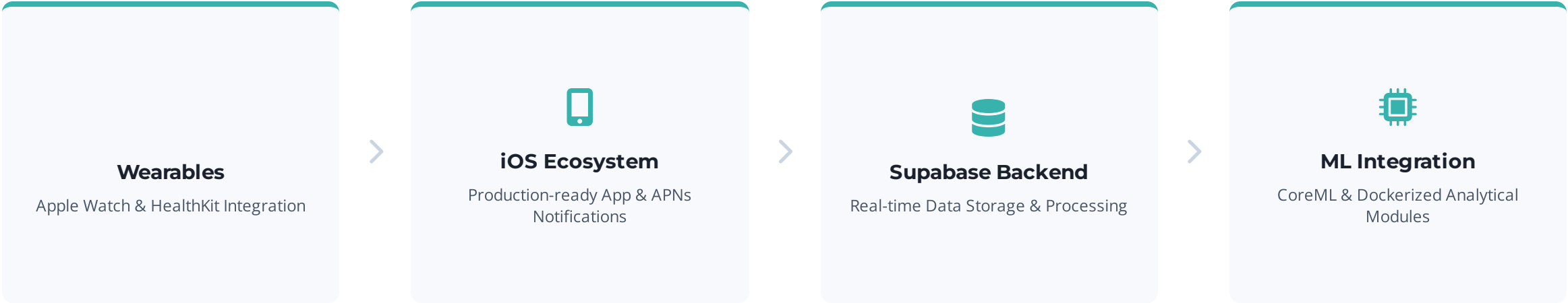


ONDA Life x TNO

R&D Collaboration for Eurostars Call 11

APPLIED SCIENCE & TECHNICAL VALIDATION • 2026

Technical Maturity: Data Science & ML Integration Ready



DATA SCIENCE READY
Architecture is fully optimized for integration with **PYTHON/R** analytical modules for advanced biometric modeling.

FLEXIBLE ML DEPLOYMENT
Support for on-device inference via **COREML** or scalable backend processing through **DOCKERIZED** containers.

EMA INFRASTRUCTURE
Ready for Ecological Momentary Assessment (EMA) via targeted push notifications and real-time biometric triggers.

HIGH-FIDELITY STREAMS
Processing of raw R-R intervals for Time-Domain and Frequency-Domain HRV (RMSSD, HF/LF) in production.

The Eurostars Innovation: Closed-Loop & Predictive Health



Predictive Behavioral Health

Transitioning from reactive tracking to **predictive modeling** of cognitive fatigue and burnout using long-term EMA data.



Closed-Loop Neuro-Sync

Developing **adaptive algorithms** that modulate audio-visual stimuli in real-time based on live HRV fluctuations.



Clinical Data Fidelity

Benchmarking commercial PPG sensors against **clinical ECG/EEG standards** to validate stress detection accuracy.

Core Goal: Elevating ONDA from a Wellness App to a Scientifically Validated Tool

Strategic Synergy: Work Packages & Legal Context

ONDA (SME Lead)

- ✓ **Architecture:** Full-stack iOS & Supabase infrastructure.
- ✓ **Deployment:** Scaling the ecosystem across international markets.
- ✓ **Consortium Lead:** Managing Eurostars Call 11 submission.

TNO (Research Partner)

- ✓ **Validation:** Designing clinical & empirical methodologies.
- ✓ **Modeling:** Co-developing closed-loop biofeedback algorithms.
- ✓ **Benchmarking:** Validating PPG data against ECG/EEG standards.

WP1: DATA PIPELINE

Architecture & Infrastructure by ONDA. Focus on EMA and real-time biometric streaming.

WP2: ALGORITHMIC MODELING

Closed-Loop Design by TNO. Developing adaptive stimuli modulation based on HRV.

WP3: CLINICAL VALIDATION

Empirical Benchmarking by TNO. Validating stress detection against medical standards.

 **Legal & Funding Context:** Project based in Asturias, Spain. Supported as an Innovative SME (SME/Autónomo) under CDTI/Sekuens criteria, ensuring national/regional alignment for Eurostars.

Collaborative Roadmap

TOWARDS A NEW STANDARD IN PREDICTIVE BEHAVIORAL HEALTH

We invite TNO to join us for a 15-minute introductory call to discuss the Eurostars Call 11 submission and explore how our technical infrastructure can support your applied research roadmap.

TECHNICAL DOCUMENTATION

onda-life.com/p/info

ONDA LIFE • 2026